Physics A (H156, H556)

Astrophysics 2
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	

Centre number						Candidate number				

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **36**.
- The total number of marks may take into account some 'either/or' question choices.

1 An early estimate for the Hubble constant was 500 km s^{-1} Mpc^{-1}.

What is the value of this estimate in units of s^{-1}?

1 parsec = 3.1×10^{16} m

A 2.3×10^{-18}
B 1.6×10^{-17}
C 1.6×10^{-5}
D 0.5

Your answer ☐ **[1]**

2 During the evolution of the universe there was a period of inflation.

Which forms of matter, if any, existed 10^{-10} s after the big bang?

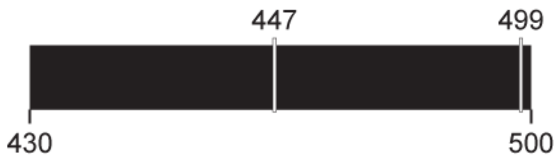
A Atoms
B Leptons
C None
D Quarks

Your answer ☐ **[1]**

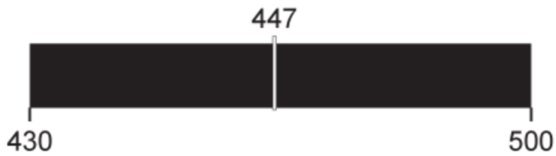
3 Part of the emission spectrum for hydrogen in a laboratory is shown. All wavelengths are given in nm.

Which diagram shows the corresponding part of the absorption spectrum observed from Earth emitted from a galaxy moving away with a velocity of 0.031 c?

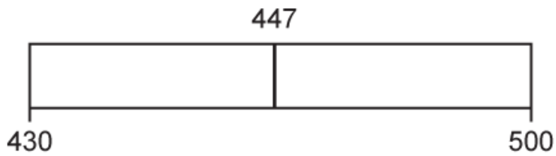
A



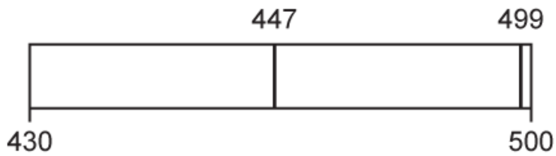
B



C

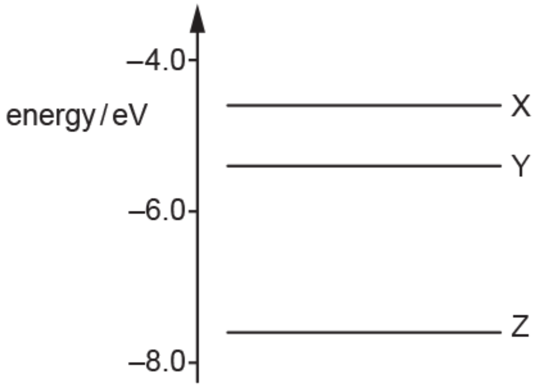


D



Your answer [1]

4 The diagram shows three energy levels X, Y and Z of an electron within a gas atom.



Which transition is correct when the electron absorbs a photon with the shortest wavelength?

A Z → X

B X → Z

C Y → X

D X → Y

Your answer [1]

5 Light from a hydrogen source is incident normally at a diffraction grating. The first order maximum of the H-alpha spectral line of wavelength 486 nm is observed at angle of 30.0°.

Light from a distant receding star is observed using the same diffraction grating. The light is incident normally at the grating as before. The speed of this star is 0.16*c*, where *c* is the speed of light in a vacuum.

What is the observed angle of the first order maximum of the H-alpha spectral line from the light of this receding star?

A 24.8°

B 30.0°

C 34.8°

D 35.5°

Your answer [1]

6 A galaxy, 1.0×10^9 light-years away from the Earth, has a recession speed of 23 000 kms^{-1}.

Which expression, based on the information above, is correct for the age of the universe in seconds?

A $\text{age} = \dfrac{1.0 \times 10^9}{23000 \times 10^3}$

B $\text{age} = \dfrac{1.0 \times 10^9 \times 1.5 \times 10^{11}}{23000}$

C $\text{age} = \dfrac{1.0 \times 10^9 \times 9.5 \times 10^{15}}{23000 \times 10^3}$

D $\text{age} = \dfrac{1.0 \times 10^9 \times 3.1 \times 10^{16}}{23000 \times 10^3}$

Your answer ☐ **[1]**

7 Astronomers observe approximately the same number of distant galaxies per unit volume of space in all directions.

Which idea does this observation support?

A Big bang model of the universe

B Cosmological principle

C Existence of dark matter

D Hubble's law

Your answer ☐ **[1]**

8 The Hipparcos space telescope used stellar parallax with a precision of 9.7×10^{-4} arcseconds to determine the distance to stars.

One of the stars studied was Polaris A. Data about this star is in the table below.

Parallax angle	7.5×10^{-3} arcseconds
Radius	2.1×10^{10} m
Mass	1.1×10^{31} kg
Surface temperature	6000 K
Temperature of the atmosphere of the star	4.0×10^{6} K

i. Estimate the maximum stellar distance in parsecs that could be measured using Hipparcos.

maximum stellar distance = ...pc **[1]**

ii. Calculate the percentage uncertainty in the calculated value of the distance to Polaris A.

percentage uncertainty = ... % **[2]**

9 A star has a mass similar to that of the Sun.

Describe how the position of this star on a Hertzsprung-Russell (H-R) diagram changes as it evolves.

Fig, A is a blank H-R diagram.

You may add information to Fig, A as part of your response.

Fig, B shows the relative intensities of different wavelengths of light in the spectrum of a star.

Explain how information from Fig. B could be used to suggest the stage of evolution of the star. Describe the limitations of the analysis.

Fig. A

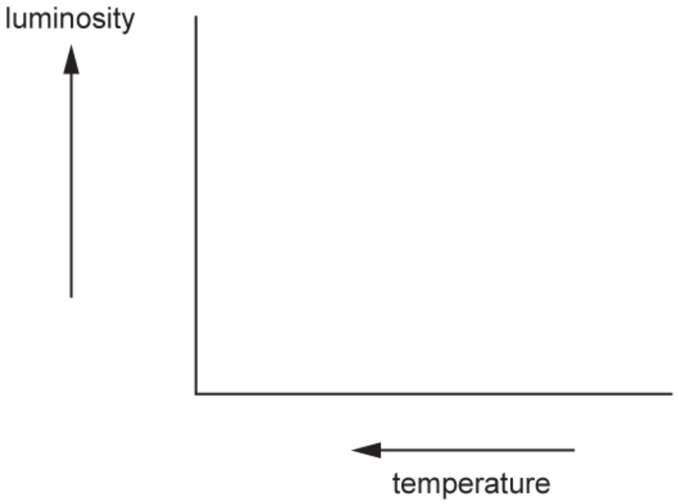
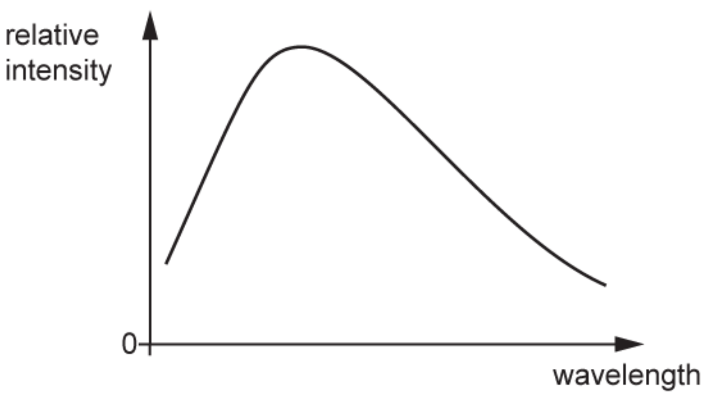


Fig. B



[6]

10(a)

A nebula is a giant cloud of gas and dust in space. A nebula X is modelled as a sphere of gas and dust particles of diameter 6.4 pc.

The nebula has 1.0×10^{12} gas and dust particles per m^3 and a temperature of 250 K. The nebula behaves like an ideal gas.

i. Show that the volume of the nebula is 4.1×10^{51} m^3.

$1 \text{ pc} = 3.1 \times 10^{16}$ m

[2]

ii. Calculate the **total** kinetic energy E_k of the gas and dust particles in the nebula.

E_k= .... J **[3]**

(b) The nebula that formed the Sun is estimated to have a diameter of 3.0 pc and had a similar composition to nebula **X** in **(b)**.

The mass of the nebula **X** is **much greate r** than the mass of the Sun.

i. Calculate the ratio $\frac{\text{mass of nebula } \mathbf{X}}{\text{mass of the Sun}}$

ratio = ... **[2]**

ii. After a long time, nebula **X** will form a stable star.

Describe the eventual evolution of this star.

[4]

11(a)

A team of astronomers have measurements to determine the peak surface temperature T and luminosity L of a distant star. They plan to use Stefan's law to estimate the radius r of this star.

Explain whether the astronomers should attempt to measure T or L more precisely to reduce the uncertainty in r.

---

[2]

(b) *It is suggested that the luminosity L and the mass M of a star can be compared to the Sun by the equation

$$\frac{L}{L_\odot} = \left(\frac{M}{M_\odot}\right)^b$$

where $L_\odot$ is the luminosity of the Sun and $M_\odot$ is the mass of the Sun.
The value of b is between 3 and 4.

Table 22 shows some data of five stars.

Main sequence star	$\frac{M}{M_\odot}$	$\frac{L}{L_\odot}$
Pi Andromedae A	6.5	800
Alpha Coronae Borealis A	3.2	80
Gamma Virginis	1.7	6.0
Eta Arietis	1.3	2.5
70 Ophiuchi A	0.78	0.4

Table 22

Fig. 2 2 shows the $\lg\left(\frac{L}{L_\odot}\right)$ against $\lg\left(\frac{M}{M_\odot}\right)$ plot for these stars.

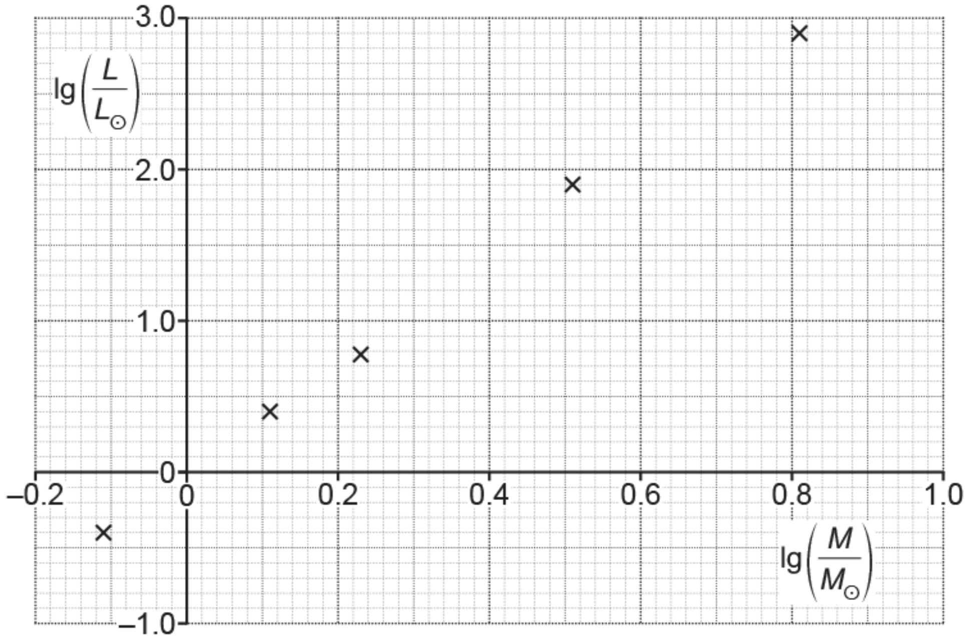


Fig. 22

The luminosity of a star is directly proportional to the rate of fusion of hydrogen nuclei.

Use **Fig. 22** to determine *b* and use your knowledge of Hertzsprung–Russell (HR) diagrams to deduce how the lifespan of hotter stars compares with lifespans of cooler stars.

[6]

12

A nebula is a giant cloud of gas and dust in space. The nebula can produce a star over a long period of time.

State what causes the initial collapse of the nebula.

[1]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			B	1	
			Total	1	
2			C	1	
			Total	1	
3			C	1	<u>Examiner's Comments</u> The spectrum in the question and those for answers A and B depict emission spectra. Spectra C and D are absorption spectra. The question asks for the absorption spectrum from distant galaxies so only answers C and D could be correct. With the recession velocity of 0.031*c*, both wavelengths will be increased by 3.1 percent. The laboratory wavelength of 434 nm becomes 447 nm after the redshift. The laboratory wavelength of 486 nm becomes 501 nm and so lies outside the range for these diagrams. This makes the correct answer C.
			Total	1	
4			A	1	<u>Examiner's Comments</u> Approximately half of all candidates chose option B, possibly because they missed the idea of photon absorption in the question. The correct answer is A, because the electron is going up the energy scale.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
5			D	1	<u>Examiner's Comments</u> Candidates found this question challenging because it joins two ideas together. The first idea is that of the Doppler effect on waves from a moving source, which will increase the wavelength received by the observer. This removes options A and B. Once the new wavelength has been calculated, the corresponding angle should be calculated from the diffraction grating equation. It is worth mentioning that the correct answer and the most likely incorrect answer are too close together for the candidate to merely guess.
			Total	1	
6			C	1	
			Total	1	
7			B	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
8		i	$=1 \times 0.00097$ =1000 parsecs	A1	1030 to 3sf
		ii	$=0.00097 \div 7.5 \times 10^{-3}$ (x100%) =13%	C1 A1	NOT half of the precision here (reference specification) Accept 1 sig fig value 10% Unrounded answer is 12.93... % <u>Examiner's Comments</u> The maximum stellar distance is given by the smallest parallax Hipparcos can measure, i.e. 9.7×10^{-4} arcsec. The distance corresponding to this parallax is $1/9.7 \times 10^{-4}$ = 1030 pc. To find the percentage uncertainty in the distance, candidates needed to divide the smallest detectable change in the parallax by the parallax itself. This is equivalent to the percentage uncertainty in the distance because of the reciprocal relationship of distance with parallax. Many candidates correctly determined both quantities. A minority of candidates confused the two similar sounding quantities.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
9			**Level 3 (5–6 marks)** Clear description, explanation and limitations with correct annotations on diagram *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Some description of evolution **and** some explanation of analysis or Some description of evolution **and** some limitations *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Limited description **or** Limited explanation or limitations *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 marks** *No response or no response worthy of credit.*	B1 x 6	Use level of response annotations in RM Assessor **Indicative scientific points may include:** **Description of evolution** • Main sequence labelled • Red giants/supergiants labelled • White dwarves labelled • Correct order of evolution • Dependence on mass/Chandrasekhar's limit • Evolutionary track shown on diagram **Explanation of analysis** • Identification of peak wavelength from graph • Use of Wien's law • Determination of temperature • Gives the horizontal co-ordinate **Limitations** • Does not give luminosity data • Cannot distinguish between similar temperature stars (e.g. white dwarf, main sequence or a hot super giant) • Need luminosity data to classify the star • Difficult to isolate light from stars in other galaxies to analyse • If looking at stars in other galaxies, have to account for red shift requiring emission data. <u>Examiner's Comments</u> Most Level of Response (LoR) questions have multiple parts to them. The first task the candidates are asked is to describe the life of a star similar to the Sun with the added complexity of showing or describing how that affects the star's position on the H-R diagram. Many candidates enjoyed considerable success here. Fig 22.2 proved more challenging to interpret. As there are no scales on the axes, it is difficult to say what colour this star is, however λ_{max} is relatively low, which points in the direction of early-mid-life (yellow or yellow white). Candidates rightly suggested that Wien's law was

Question			Answer/Indicative content	Marks	Guidance
					applicable and that as the star grew older and expanded into a red giant, the curve's peak would move to the right. The most challenging aspect of this question is suggesting what limitations this approach has. The lack of scales has already been mentioned. The colour alone cannot uniquely define a star's position on the H-R diagram and thus the stage of life of the star. The graph only gives a relative intensity, which means that the total output of the star, or its luminosity cannot be measured. Exemplar 3

Fig. 22.1

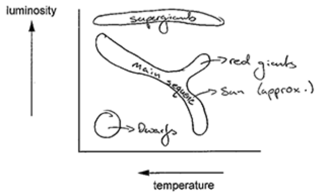
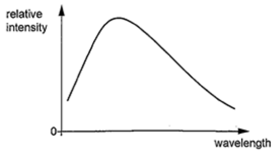


Fig. 22.2



When the star begins fusing, becoming a star from ~~the~~ a protostar in the nebula cloud it will be in the bottom right of the diagram and progress to around the position of the sun as it will be

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Question			Answer/Indicative content	Marks	Guidance
					relatively low temperature and luminosity. Once it has fused all of its hydrogen, the unfusing core will collapse and the outer layers will expand and cool forming a red giant due to the radiation pressure of the fusing nuclei caused by the shockwaves in the collapse. The star won't get any hotter but surface area and $\therefore$ luminosity will increase, shifting its position up and right to the red giant section. The outer layers drift off as planetary nebula into space leaving a blue dwarf which will cool to a black dwarf. Additional space if required The core will collapse until electron degeneracy pressure matches gravitational collapse. It's surface area is much smaller now $\therefore$ luminosity is low but temperature is high due to gravitational collapse so it shifts to the bottom left of the diagram where white dwarves show. Wein's displacement law can be used to calculate the temperature of a star ($\lambda \propto 1/T$) which can then be used to locate its "x-coordinate" on the HR diagram. However we don't know the star's surface area so luminosity is unknown and so it can be hard to tell apart a white dwarf or high luminosity main sequence star for example as they are the same temperature. This candidate described the life cycle of sun-like stars in great depth, completed an excellent HR diagram and in the text described the path of the star on the H-R diagram throughout its lifetime. The references in the last paragraph are what makes this response worthy of a Level 3. The candidate has answered all aspects of the question, explaining both Wien's law and how the luminosity (and hence position on the H-R diagram) is impossible from the data presented.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
10	a	i	(diameter =) $6.4 \times 3.1 \times 10^{16}$ **or** 2.0×10^{17} (m) (volume =) $4/3\ \pi \times (9.9 \times 10^{16})^3$ (volume =) 4.1×10^{51} (m^3)	C1 C1 A0	**Allow** (radius =) $3.2 \times 3.1 \times 10^{16}$ **or** 9.9×10^{16} (m) <u>Examiner's Comments</u> Candidates successfully converted the radius from parsecs into metres and from there the volume of the nebula.
		ii	$(E = \frac{3}{2}kT)\ \frac{3}{2} \times 1.38 \times 10^{-23} \times 250$ **or** 5.2×10^{-21} (J) (N =) $1.0 \times 10^{12} \times 4.1 \times 10^{51}$ **or** 4.1×10^{63} ($E_k = 4.1 \times 10^{63} \times 5.2 \times 10^{-21}$) $E_k = 2.1 \times 10^{43}$ (J)	C1 C1 A1	<u>Examiner's Comments</u> This brief yet multi-stage question proved relatively challenging. The correct approach here was to find the average kinetic energy of a single particle (using Ek = 3/2 kT) and then multiplying this by the number of particles in the nebula. The number of particles in the nebula was found by multiplying the number density by the volume of the nebula.
	b	i	Mass is proportional to volume **or** diameter^3 **or** radius^3 **or** $(\frac{6.4}{3})^3$ **or** $(\frac{3.2}{1.5})^3$ ratio = 9.7	C1 A1	**Allow** attempt at calculating volume of second nebula and comparing volumes directly **Allow** 9.76 (if volume divided by volume of Sun's nebula) <u>Examiner's Comments</u> Candidates used several different approaches here. By assuming a similar density, the mass is directly proportional to the volume. Some candidates calculated the volume of the Sun's nebula. Others correctly assumed that the volume and hence mass of each nebula was directly proportional to the diameter^3 (or radius^3).

Question			Answer/Indicative content	Marks	Guidance
		ii	Fuel (hydrogen) runs out Super red giant star (Mass of core > Chandrasekhar limit /1.4 therefore) supernova neutron star or black hole (formed)	B1 B1 × 3	**Note: incorrect order is CON** **Allow** alternative route: Red giant formed (mass of star< 10 solar masses, therefore) planetary nebula (and) white dwarf formed <u>Examiner's Comments</u> Some candidates used the ratio from the previous part of the question to assume that the star from nebula X would eventually become a white dwarf, as suggested in one of the endorsed textbooks. Others used the information in the question, i.e. that the mass of nebula X was far greater than that of the Sun. This meant they were justified in assuming that this particular star would become a supernova. Both approaches were acceptable, provided the candidate chose one route and described it correctly.
			Total	11	

Question			Answer/Indicative content	Marks	Guidance
11	a		(%) uncertainty of *T* will be 4 times as significant as the uncertainty of *L* Should improve measurements leading to *T*	M1 A1	**Allow** reference to <u>4</u>th power of T **Allow** comparison of $L^{(-)1/2}$ and T^2 <u>Examiner's Comments</u> As there is a choice of only two variables here, the first mark was necessarily a 'method' mark. The best answers stated that Stefan's law uses T^4 and so the percentage uncertainty in T would be 4 times more significant (as described in the practical guidance). **OCR support** Teachers and students all have access to the Practical Skills handbook which includes techniques and ideas that prove useful in answering Level of Response and other questions that check practical experience.

Question			Answer/Indicative content	Marks	Guidance
	b		***Level 3 (5–6 marks)** Clear description of method and analysis of data **and** correct explanation. *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Some description of method **and** analysis of data **or** explanation. *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Limited description **or** Limited analysis **or** Limited explanation *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 marks** *No response or no response worthy of credit.*	B1 × 6	**Indicative scientific points may include**: **Description of method** • Equation using lg of both sides • Use of the lg $X^b = b \lg X$ • Comparison with $y = mx$ (+ c) • (y-intercept = 0) **Analysis of data** • Straight line through the origin • Gradient of the graph is b • Gradient calculated to be between 3 and 4 **Explanation** • Labelled sketch of HR diagram • Reference to Stefan's Law • Hotter stars (than the Sun) have greater luminosity (ratio) / luminosity ratio >1 • Hotter stars have much smaller mass ratio than luminosity ratio • Use of data (e.g luminosity of ×10,000 means mass of about ×14) • **therefore** hotter stars lose mass at a much higher rate (compared to their mass) • **therefore** hotter star lifespans are very much shorter than cooler stars <u>Examiner's Comments</u> This question relied on a small number of different skills. At first the candidate needed to understand how to find the constant '*b*' and be able to communicate that. Comparison with '$y = mx + c$' invariably helps with this task. Once '*b*' was determined, most candidates didn't refer to it in the rest of their argument, limiting themselves to a Level 2 response. Instead, they talked about features of the HR diagram rather than focusing on the relevance of '*b*'. '*b*' is important because it makes it clear that a small increase in mass gives a much larger increase in luminosity. Luminosity is related to the mass loss per second so a larger luminosity means the mass of hydrogen will 'run out' far quicker in comparison to a larger star.

Question			Answer/Indicative content	Marks	Guidance
			Total	8	
12			Gravitational force	B1	**Allow** ‘gravity’ <u>Examiner’s Comments</u> Most candidates mentioned a word that showed they knew the collapse was to do with gravitational self-attraction.
			Total	1	